DairySense Database Schema

This document outlines the entire PostgreSQL database architecture for DairySense. For your meeting, you can use the interactive ER diagram or the module breakdowns below.

🔗 Entity-Relationship Diagram

```
erDiagram
    users {
        int id PK
        string email UK
        string password_hash
        timestamp created_at
    }

    cows {
        int id PK
        string cow_id UK
        string rfid_uid UK
        string cow_tag UK
        string mother_id
        string parent_id
        string name
        string cow_type "normal, pregnant, dry, calf, milking"
        string breed
        int weight_kg
        string source_type
        date date_of_birth
        string status
        boolean is_active
    }

    daily_lane_log {
        int id PK
        date date
        int lane_no
        string cow_id FK
        string cow_type
        decimal feed_given_kg
        decimal morning_yield_l
        decimal evening_yield_l
        decimal total_yield_l
    }

    milk_yield_log {
        int id PK
        date date
        string cow_id FK
        string session "Morning, Evening"
        decimal milk_qty_kg
```

```
    decimal milk_qty_litre
}

feed_category_master {
    int id PK
    string category_name UK
}

feed_item_master {
    int id PK
    int category_id FK
    string item_name
    string default_unit
}

feed_log {
    int id PK
    date date
    int feed_item_id FK
    decimal quantity_kg
    decimal cost_per_unit
    decimal total_amount
    string input_source "Purchased, Farm Produced"
}

medicine_master {
    int id PK
    string medicine_name UK
    string category "Medicine, Supplement, Treatment"
}

cow_medicine_log {
    int id PK
    string cow_id FK
    int medicine_id FK
    date date_given
    string dosage
    string administered_by
}

cow_stage_master {
    int id PK
    string stage_name UK
    string stage_type
}

activity_master {
    int id PK
    string activity_name
    int stage_id FK
    string frequency
}
```

```
cow_activity_schedule {
    int id PK
    string cow_id FK
    int activity_id FK
    date due_date
    string status "Pending, Completed, Skipped"
}

daily_cow_metrics {
    int id PK
    string cow_id FK
    date date
    decimal feed_given_kg
    decimal milk_yield_litre
}

daily_farm_summary {
    int id PK
    date date UK
    decimal total_feed_kg
    decimal total_milk_litre
    string best_cow_id
    string lowest_cow_id
}

cows ||--o{ daily_lane_log : "logs daily"
cows ||--o{ milk_yield_log : "yields"
cows ||--o{ cow_medicine_log : "receives"
cows ||--o{ cow_activity_schedule : "has scheduled"
cows ||--o{ daily_cow_metrics : "metrics"
feed_category_master ||--|{ feed_item_master : "categorizes"
feed_item_master ||--o{ feed_log : "records usage"
medicine_master ||--o{ cow_medicine_log : "prescribed in"
cow_stage_master ||--o{ activity_master : "defines"
activity_master ||--o{ cow_activity_schedule : "triggers"
```

🪜 Core Modules Breakdowns

1. Authentication & Users

- **`users`** : Secure backend table handling dashboard access. Tracks `email` (unique) and `password_hash` .

2. Cattle Management

- **`cows`** : The primary master entity describing every animal on the farm.
  - Generates a unique string `cow_id` (e.g. COW-001) alongside native integer `id` .
  - Supports genetics tracking (`mother_id` , `parent_id`), identification (`cow_tag` , `rfid_uid`), profiling (`name` , `breed` , `weight_kg` , `cow_type`), and health histories.

3. Production & Logging

- **`daily_lane_log`** : The fundamental daily physical operation log. Aggregates data by unique pairs of `[date, lane_no, cow_id]` . Tracks both `feed_given_kg` and specific `morning_yield_l` / `evening_yield_l` .
- **`milk_yield_log`** : Detailed parallel logging specific strictly to milk yields per `Morning` or `Evening` session.

4. Feed & Inventory Management

- **`feed_category_master`** : Base groupings like *Concentrated Feed*, *Dry Fodder*, etc.
- **`feed_item_master`** : Individual feed types like *Hay Stack* and *Corn Flour*. Tied to their Categories.
- **`feed_log`** : Captures bulk tracking of how feed is consumed across the farm. Records exact financial and stock costs mapping to `quantity_kg` , `cost_per_unit` , `total_amount` , and where the feed came from (`input_source`).
- **`cow_weight_groups` & `feed_requirement_rules`** : Links feed consumption formulas to different cow weight profiles dynamically.

5. Veterinary & Medicine

- **`medicine_master`** : A directory detailing all farm medicines, supplements, or treatments.
- **`cow_medicine_log`** : Administered inventory logic binding a specific `medicine_id` to a `cow_id` with dosage records and who administered it.

6. Activity Sequencing

- **`cow_stage_master`** : Represents the current phase of the cow (e.g. *Pregnant Month 1*, *Lactation*, *Calf*).
- **`activity_master`** : Specific actionable tasks mapped to stages (like *Vaccination* or *Delivery Prep*).
- **`cow_activity_schedule`** : The live calendar. Populated with `Pending` , `Completed` , or `Skipped` activities mapped specifically to individual cows.

7. Performance Monitoring

- **`daily_cow_metrics`** : Summarized tracking per cow per day ensuring rapid querying times for dashboards.
- **`cow_daily_status`** : Alerting and welfare flagging table mapping `NORMAL` , `SLIGHT_DROP` , or `ATTENTION` states triggered by AI formulas.
- **`daily_farm_summary`** : Entire farm overview aggregations by `date` (Total Feed, Best Earner, Lowest Earner).